

KINEMATICS

The Complete Comprehensive Reference & Lecture Notes

Subject: Theoretical and Mathematical Physics

Level: Advanced Undergraduate / Competitive Physics Module

Scope: Complete Analysis of Motion in 1D, 2D, Projectiles, Vectors, and Circular Systems

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1. CLASSICAL MECHANICS AND FOUNDATIONS OF KINEMATICS

Mechanics is the foundational branch of physical science that investigates the state of rest or motion of material bodies subjected to forces or displacements. Structurally, the discipline is split into three core subsections: Statics, Kinematics, and Dynamics.

1.1 Categorization of Mechanics

- **Statics:** The study of material bodies at rest or moving with a constant velocity such that the net acceleration of the system is identically zero, existing under a state of static equilibrium.
- **Kinematics:** Derived from the Greek term "*kinema*" (meaning motion). It serves as the purely geometry-driven description of the motion of particles, objects, or systems of bodies without any regard for the underlying physical causes, mass characteristics, or driving forces that precipitate such motion.
- **Dynamics:** The study of the mathematical and physical relationship between the moving body and its underlying causal mechanisms, primarily encompassing forces, inertia, momentum, energy, and torques.

1.2 The Relativity of Space: Rest and Motion

A fundamental axiom of classical physics is that absolute rest or absolute motion does not exist within the universe. Every physical state is inherently relative and requires the selection of a coordinate anchor point. An object is defined to be in a state of motion if its spatial coordinates vary continuously over a finite temporal sequence with respect to a designated observer. Conversely, if the spatial coordinates remain completely invariant throughout a given temporal sequence, the object is defined to be at rest.

Axiomatic Framework: Consider a passenger seated inside a train moving with a uniform velocity vector $\vec{V}$ with respect to the earth's surface. To another passenger sitting directly across, the target passenger's positional coordinates do not evolve over time, implying a state of mechanical rest. However, to a stationary observer standing on the track platform, the spatial coordinates of the same passenger are changing linearly, establishing a state of active translation. Hence, the properties of rest and motion are completely dictated by the selection of the observer's spatial coordinates.

1.3 Mathematical Definition of a Frame of Reference

To mathematically characterize the position and movement of an arbitrary particle, a physical observer must construct a **Frame of Reference**. This system consists of an established Cartesian Coordinate System

possessing three mutually orthogonal spatial axes ($\mathbf{X}$, $\mathbf{Y}$, $\mathbf{Z}$) intersecting at a singular origin point ($\mathbf{O}$), tightly coupled with an idealized synchronized clock to measure time increments (t).

[DIAGRAM 1.1: Standard 3D Cartesian Coordinate Reference System showing orthogonal axes X , Y , and Z intersecting at Origin $O(0,0,0)$ with an absolute time clock ' t ' monitoring a moving point P with spatial vector $r(t)$]

Frames of reference are systematically classified into two categories based on their acceleration state:

1. **Inertial Frames of Reference:** Reference systems that are either completely stationary or undergoing linear translation at a perfectly uniform velocity vector. Within these frames, Newton's first law of motion remains perfectly valid without modification.
2. **Non-Inertial Frames of Reference:** Reference systems that undergo a finite linear or angular acceleration with respect to an inertial anchor. To apply standard classical mechanics within these systems, an observer must introduce a mathematical correction known as a pseudo-force or inertial force.

2. KINEMATICS OF A PARTICLE IN ONE DIMENSION

One-Dimensional or Rectilinear Motion describes the physical state where a moving object is strictly confined to a single linear path. Consequently, the position of the target particle can be completely mapped over time using a singular scalar function $x(t)$ mapped along a single coordinate dimension.

2.1 Positional Vector Analysis: Distance versus Displacement

Understanding the exact geometric properties of a particle path requires an analytical separation between scalar path length and vector spatial shifts.

Analytical Attribute	Distance (s)	Displacement (Δx)
Fundamental Nature	Scalar quantity containing only magnitude information.	Vector quantity explicitly combining magnitude and direction.
Path Dependency	Completely dependent on the actual geometric path traversed.	Independent of path topology; relies strictly on endpoints.
Mathematical Sign	Strictly monotonic non-negative value ($s \geq 0$).	Can be positive, negative, or zero based on axis direction.
Geometric Inequality	$Distance \geq Displacement $	

[DIAGRAM 2.1: Particle path tracking from Point A to Point B. The irregular winding line indicates total scalar path length (Distance), whereas the direct, straight-line arrow vector directly connecting A to B indicates the Displacement Vector]

Let the initial state of a point mass be denoted by the position coordinate x_1 at time t_1 , and its subsequent state be denoted by coordinate x_2 at a later time t_2 . The net mathematical displacement vector Δx during this interval is explicitly written as:

$$\Delta x = x_2 - x_1$$

2.2 Temporal Rates of Positional Change: Speed and Velocity

2.2.1 Average and Instantaneous Speed

Average speed is defined as the total scalar distance covered by a particle divided by the entire elapsed time interval. It provides no information regarding localized variations during translation.

$$\text{Speed}_{avg} = \text{Total Distance} / \text{Total Time} = s / \Delta t$$

Instantaneous speed represents the exact magnitude of the spatial rate of change of distance evaluated at an infinitely small temporal window. Mathematically, it is the first derivative of the distance function:

$$\text{Speed}_{inst} = \lim_{\Delta t \rightarrow 0} (\Delta s / \Delta t) = ds / dt$$

2.2.2 Vector Velocity Formulations

Velocity evaluates the rate of change of the displacement vector over time. The average velocity vector is computed as the net displacement divided by the time interval:

$$\mathbf{v}_{avg} = \Delta \mathbf{x} / \Delta t = (\mathbf{x}_2 - \mathbf{x}_1) / (t_2 - t_1)$$

To acquire the exact velocity at a single instant, the time step must approach zero as a limit. This converts the average parameter into a differential derivative:

$$\mathbf{v}(t) = \lim_{\Delta t \rightarrow 0} (\Delta \mathbf{x} / \Delta t) = d\mathbf{x} / dt$$

2.3 Acceleration Dynamics

Acceleration characterizes the formal time rate of change of the velocity vector. If a particle's velocity vector varies either in terms of its direct magnitude or its spatial orientation, the particle is accelerating.

$$\mathbf{a} = d\mathbf{v} / dt = d^2\mathbf{x} / dt^2$$

Using the mathematical chain rule of calculus, acceleration can also be formulated without an explicit time parameter by relating it directly to spatial coordinates:

$$a = (dv / dx) \cdot (dx / dt) = v \cdot (dv / dx)$$

3. RIGOROUS DERIVATION OF KINEMATIC EQUATIONS

When a particle moves along a rectilinear path under a perfectly constant or uniform acceleration field ($a = \text{constant}$), its motion parameters can be completely solved using algebraic links. We will derive these expressions using calculus principles.

3.1 Velocity-Time Relation (First Equation of Motion)

By definition, instantaneous acceleration is given by the differential expression:

$$a = dv / dt \quad dv = a \cdot dt$$

To compute the overall velocity change, we integrate both sides of the differential expression. We apply boundaries where at time $t = 0$, initial velocity equals u , and at time $t = t$, final velocity equals v :

$$\int_u^v dv = \int_0^t a \, dt$$

Since acceleration a is constant, it can be factored out of the integral block:

$$\begin{aligned} [v]_u^v &= a [t]_0^t \\ v - u &= a(t - 0) \\ v &= u + at \end{aligned}$$

3.2 Displacement-Time Relation (Second Equation of Motion)

Velocity is defined as the first derivative of spatial position with respect to time:

$$v = ds / dt \quad ds = v \cdot dt$$

Substituting the previously derived expression for velocity ($v = u + at$) into this equation yields:

$$ds = (u + at) \, dt$$

We execute definite integration across explicit physical limits. At $t = 0$, the scalar displacement is $s = 0$, and at $t = t$, total displacement reaches s :

$$\begin{aligned}\int_0^s ds &= \int_0^t (u + at) dt \\ s &= \int_0^t u dt + \int_0^t at dt \\ s &= u [t]_0^t + a [\frac{1}{2}t^2]_0^t \\ s &= ut + \frac{1}{2}at^2\end{aligned}$$

3.3 Velocity-Displacement Relation (Third Equation of Motion)

Using the spatial form of acceleration derived via the calculus chain rule:

$$a = v \cdot (dv / ds) \quad a ds = v dv$$

Integrating across explicit physical boundaries where at $s = 0$, velocity is u , and at position s , velocity is v :

$$\begin{aligned}\int_0^s a ds &= \int_u^v v dv \\ a [s]_0^s &= [\frac{1}{2}v^2]_u^v \\ as &= \frac{1}{2}(v^2 - u^2) \\ v^2 &= u^2 + 2as\end{aligned}$$

3.4 Displacement in a Specific n^{th} Second Interval

To compute the displacement covered during a precise single second interval ending at time $t = n$, we must subtract the total displacement covered in $(n-1)$ seconds from the total displacement covered in n seconds:

$$s_n^{\text{th}} = s_n - s_{n-1}$$

Applying the second equation of motion for both instances:

$$\begin{aligned}s_n^{\text{th}} &= [un + \frac{1}{2}an^2] - [u(n-1) + \frac{1}{2}a(n-1)^2] \\ s_n^{\text{th}} &= un + \frac{1}{2}an^2 - un + u - \frac{1}{2}a(n^2 - 2n + 1) \\ s_n^{\text{th}} &= u + a/2 \cdot (2n - 1)\end{aligned}$$

4. MOTION UNDER THE INFLUENCE OF GRAVITY (FREE FALL)

A ubiquitous manifestation of constant rectilinear acceleration is the vertical motion of bodies within proximity of the Earth's surface. Minimizing atmospheric drag forces yields an idealized state known as **Free Fall**, where

the constant downward gravitational acceleration is denoted as $g \approx 9.81 \text{ m/s}^2$.

4.1 Sign Convention Standards

To prevent vector orientation errors, a standard coordinate convention must be enforced throughout the calculations:

- Choose the point of release as the absolute coordinate origin $(0,0)$.
- Vectors pointing vertically upward are assigned a positive mathematical sign (+).
- Vectors pointing vertically downward are assigned a negative mathematical sign (-).
- Consequently, the acceleration parameter is always negative: $a = -g$.

4.2 Comprehensive Case Analyses

Case Analysis I: A particle dropped from a stationary height H ($u = 0$)

Setting the initial position at height H , the landing point corresponds to a total vertical translation of $s = -H$. Applying these parameters to the second equation of motion:

$$-H = (0)t + \frac{1}{2}(-g)t^2 \quad H = \frac{1}{2}gt^2 \quad t_{\text{fall}} = \sqrt{(2H/g)}$$

To compute the impact velocity just prior to hitting the ground, we apply the parameters to the third equation of motion:

$$v^2 = 0^2 + 2(-g)(-H) \quad v = \sqrt{(2gH)}$$

Case Analysis II: A particle projected vertically upward with initial speed u

At the peak of its ascent, the particle's instantaneous velocity momentarily drops to zero ($v = 0$). To compute the peak height reached (H_{max}):

$$0 = u^2 + 2(-g)(H_{max}) \quad H_{max} = u^2 / (2g)$$

The total time required to reach this peak (Time of Ascent) is derived using the first equation:

$$0 = u + (-g)t_{up} \quad t_{up} = u / g$$

Due to the symmetric nature of conservative gravitational fields, the time of ascent equals the time of descent, yielding a total time of flight of:

$$T_{total} = 2u / g$$

5. ADVANCED GRAPHICAL REPRESENTATION OF KINEMATIC STATES

Graphical calculus serves as an important visual method for diagnosing and analyzing a particle's kinematic states. Slopes and integrated areas contain explicit physical definitions.

5.1 Calculus Meanings of Curves

1. **Position-Time (\$x\$-\$t\$) Plots:** The instantaneous tangent slope evaluated at any point along an x - t curve maps directly to the particle's instantaneous velocity:

$$\text{Slope} = dx / dt = v$$

2. **Velocity-Time (\$v\$-\$t\$) Plots:**

- The instantaneous tangent slope maps directly to acceleration: $\text{Slope} = dv/dt = a$.
- The definite integral (the area enclosed between the curve and the time axis) maps directly to the net displacement:

$$\text{Displacement} = \int v \, dt = \text{Area}$$

3. **Acceleration-Time (\$a\$-\$t\$) Plots:** The area enclosed beneath an a - t curve across a specified time window represents the net change in velocity:

$$\Delta v = \int a \, dt = \text{Area}$$

[DIAGRAM 5.1: Three synchronized stacked kinematic graphs representing constant acceleration. Top: Concave upward parabolic x - t plot. Middle: Linearly increasing v - t plot highlighting the tangent angle. Bottom: Flat, constant positive horizontal a - t line]

6. KINEMATICS OF A PARTICLE IN TWO DIMENSIONS:

PROJECTILE MOTION

When a particle moves in a two-dimensional plane, its position vector must be separated into independent horizontal and vertical coordinate axes. The classic application of this methodology is **Projectile Motion**.

6.1 Principle of Independent Orthogonal Axes

Galileo's principle of superposition asserts that the horizontal and vertical motions of a projected particle operate completely independently of one another. The horizontal motion experiences zero net force and zero acceleration, while the vertical motion is continuously governed by the constant downward acceleration of gravity.

[DIAGRAM 6.1: Full parabolic trajectory of a projectile launched from origin with velocity u at an angle θ . Points along the path show the unchanging horizontal velocity vector component ($u \cos \theta$) alongside the continuously decreasing, zeroing, and reversing vertical component vector ($u \sin \theta$)]

6.2 Comprehensive Trajectory Mathematics

Consider a particle launched from an origin $(0,0)$ with an initial velocity magnitude u oriented at an angle θ relative to the horizontal plane.

Horizontal Component Analysis:

$$u_x = u \cos \theta ; \quad a_x = 0 ; \quad x(t) = (u \cos \theta)t$$

Vertical Component Analysis:

$$u_y = u \sin \theta ; \quad a_y = -g ; \quad y(t) = (u \sin \theta)t - \frac{1}{2}gt^2$$

6.2.1 Derivation of the Trajectory Equation

To compute the explicit spatial path equation, we solve for the time parameter t using the horizontal position equation and substitute it into the vertical position equation:

$$t = x / (u \cos \theta)$$

Substituting this expression for t yields the mathematical equation of a parabola:

$$y = x \cdot [u \sin \theta / (u \cos \theta)] - \frac{1}{2}g [x / (u \cos \theta)]^2$$

$$y = x \tan \theta - [gx^2 / (2u^2 \cos^2 \theta)]$$

6.2.2 Calculation of Primary Projectile Constants

1. **Total Time of Flight (ST\$):** The time required for the projectile to return to the initial launch plane ($y = 0$):

$$0 = (u \sin \theta)T - \frac{1}{2}gT^2 \quad T = (2u \sin \theta) / g$$

2. **Maximum Altitude Reached (SH\$):** The vertical displacement achieved when the vertical velocity component drops to zero ($v = 0$): y

$$0 = (u \sin \theta)^2 - 2gH \quad H = (u^2 \sin^2 \theta) / (2g)$$

3. **Total Horizontal Range (SR\$):** The net horizontal distance covered during the entire time of flight:

$$R = u_x \cdot T = (u \cos \theta) \cdot [(2u \sin \theta) / g]$$

$$R = [u^2 \cdot (2 \sin \theta \cos \theta)] / g$$

$$R = (u^2 \sin 2\theta) / g$$

Optimization Criteria The maximum horizontal range is dictated by the trigonometric parameter $\sin 2\theta$. The maximum possible value for a sine function is 1, which occurs when its argument equals 90 degrees. Therefore, setting $2\theta = 90^\circ$ yields an optimal launch angle of $\theta = 45^\circ$.

7. RELATIVE KINEMATICS AND MULTI-BODY FRAMEWORKS

Relative kinematics investigates the mathematical translation of motion parameters between distinct moving frames of reference.

7.1 Mathematical Definition of Relative Vector Velocity

Let two individual bodies, A and B , be moving through space with absolute velocity vectors $\vec{v}_A$ and $\vec{v}_B$ relative to a stationary ground reference frame. The relative velocity vector of body A as observed from the moving frame of reference of body B is written as:

$$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$$

Similarly, the relative acceleration vector of body A with respect to body B is defined by the vector difference:

$$\vec{a}_{AB} = \vec{a}_A - \vec{a}_B$$

7.2 Classical Applied Relative Problems

- 1. Intercept and Collision Problems:** Two moving particles will collide if their relative velocity vector $\vec{v}_{AB}$ aligns directly along the line connecting their instantaneous positions.
- 2. Rain-Man Vector Systems:** A person running horizontally through vertically falling rain will observe the raindrops approaching at an inclined relative angle. To maximize protection, the observer must tilt their umbrella parallel to the relative velocity vector $\vec{v}_{rain/man} = \vec{v}_{rain} - \vec{v}_{man}$.

8. KINEMATICS OF UNIFORM CIRCULAR MOTION

When a particle moves along a circular path of constant radius r at a completely uniform scalar speed, its motion is classified as **Uniform Circular Motion**.

8.1 Angular Metric Parameterization

- **Angular Displacement ($\Delta\theta$):** The angle swept out at the center of the circular path by the moving radius vector. It is measured in radians.

$$\Delta\theta = \Delta s / r$$

- **Angular Velocity (ω):** The time rate of change of the angular displacement vector:

$$\omega = d\theta / dt = v / r$$

8.2 The Centripetal Acceleration Field

Although the scalar speed remains constant in uniform circular motion, the direction of the velocity vector changes continuously at every point along the circular path. This constant variation in direction generates a localized acceleration vector known as **Centripetal Acceleration**. This vector points inward along the radius toward the center of curvature.

$$a_c = v^2 / r = \omega^2 r$$

[DIAGRAM 8.1: Circular path of radius r showing the constant magnitude linear velocity vector tangent to the boundary curve, paired with the orthogonal centripetal acceleration vector pointing toward the center point O]